Steps 4-6: Model Training & Evaluation

GSK Medicine Prediction System | Day 2 Summary

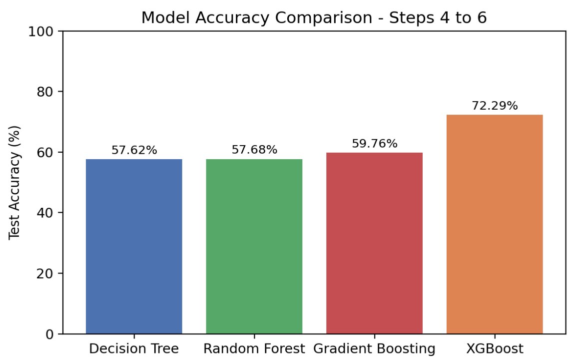
Work Completed

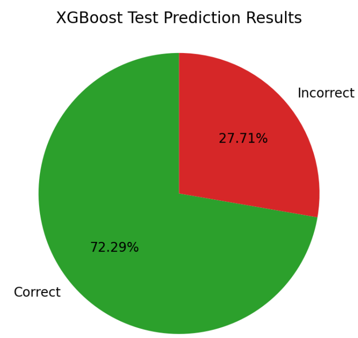
- 985,876 labelled patient records were used
- 64,124 missing-target records were excluded only for supervised model training
- The same stratified 80% training and 20% testing split with random_state=42 was used for all models
- Decision Tree, Random Forest, XGBoost and Gradient Boosting were trained
- Numeric missing values were filled using training medians
- Categorical missing values were filled using training modes and then one-hot encoded
- Preprocessing was fitted only on training data
- patient_id, treatment_outcome, admission_date, adverse_event and readmission_30d were excluded from predictors

Model Results (Test)

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Decision Tree	57.62%	29.89%	66.74%	41.29%	64.70%
Random Forest	57.68%	29.77%	65.84%	41.00%	64.47%
Gradient Boosting	59.76%	30.61%	63.28%	41.26%	65.12%
XGBoost	72.29%	36.30%	31.89%	33.95%	64.97%

Charts





Key Findings

- XGBoost achieved the highest accuracy and precision
- Decision Tree achieved the highest recall
- Gradient Boosting achieved the highest ROC-AUC
- Random Forest did not significantly improve over Decision Tree with the supplied settings
- Training and testing scores were close, indicating no serious overfitting
- Accuracy should not be considered alone because the target is imbalanced

Conclusion

Steps 4, 5 and 6 were completed successfully. XGBoost is the accuracy-leading model, while Decision Tree is strongest for identifying a larger percentage of effective-treatment cases.